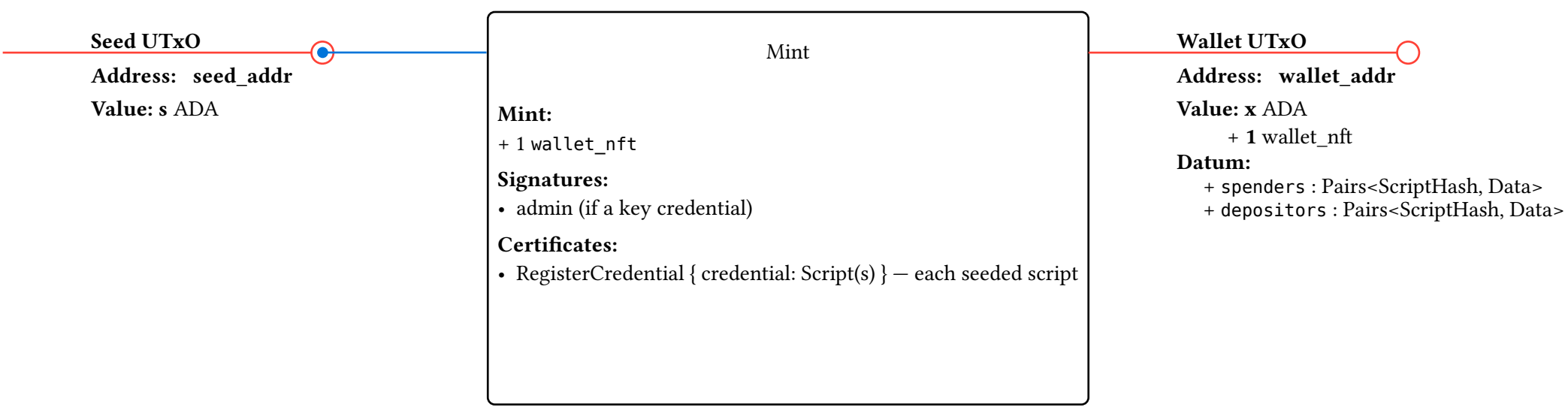
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1 - Mint (create the wallet)

The wallet is created by minting its identifying NFT — one token under the script's own policy — into a fresh wallet UTxO at the wallet address. The `admin` must authorize the mint.



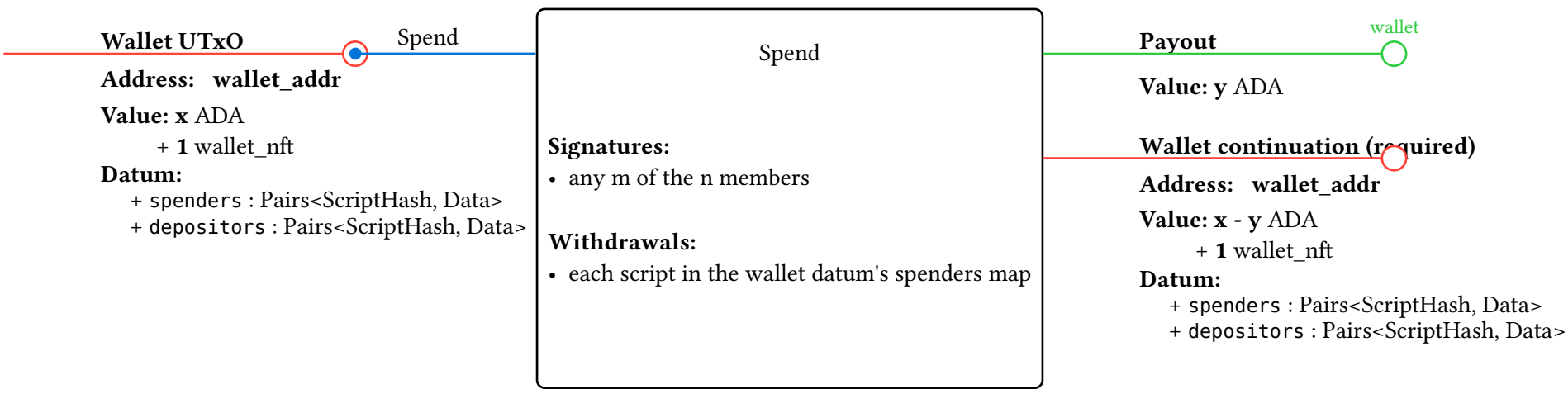
Notes:

- `wallet_addr`: payment is `wallet_hash`, staking any. `wallet_hash = hash(smart_wallet(seed_utxo, settings_ref, wallet_nft_name, admin))` — the multivalidator is parameterized by its seed, configuration reference, NFT name, and the `admin` credential, so all wallet UTxOs share one address and one NFT policy.
- one-shot: the mint requires the `seed_utxo` to be spent, and a UTxO can be spent at most once, so this policy id mints exactly one token ever. The resulting NFT is unforgeable and identifies the wallet.
- the mint redeemer is `Mint { out_ix }`, locating the new wallet UTxO.
- the new wallet UTxO carries the NFT, an inline datum with `spenders` and `depositors` maps, and no reference script. Every script in both maps must be registered in this same transaction — a `RegisterCredential` certificate for its stake credential and the matching `Publish` redeemer (the CIP-69 script witness) — which runs the script's `publish` handler. The handler reads its own initial `Data` from this output's datum and validates it, so a wallet may be created already carrying delegated restrictions or deposit validations, but only ones that self-validated their initial state.
- the `admin` credential (a script parameter) must be satisfied: a key admin signs; a script admin authorizes by a withdraw-0 invocation.

Figure 1: Mint (create the wallet)

2 - Spend (base M-of-N + delegated authorizations)

The wallet pays out. The spending script enforces its base M-of-N signature floor and forwards every other check to the withdrawal scripts named in the wallet UTxO's datum; the single spend redeemer asserts their conjunction. The wallet must survive the spend: an NFT-bearing continuation carries `depositors` *forward unchanged.*



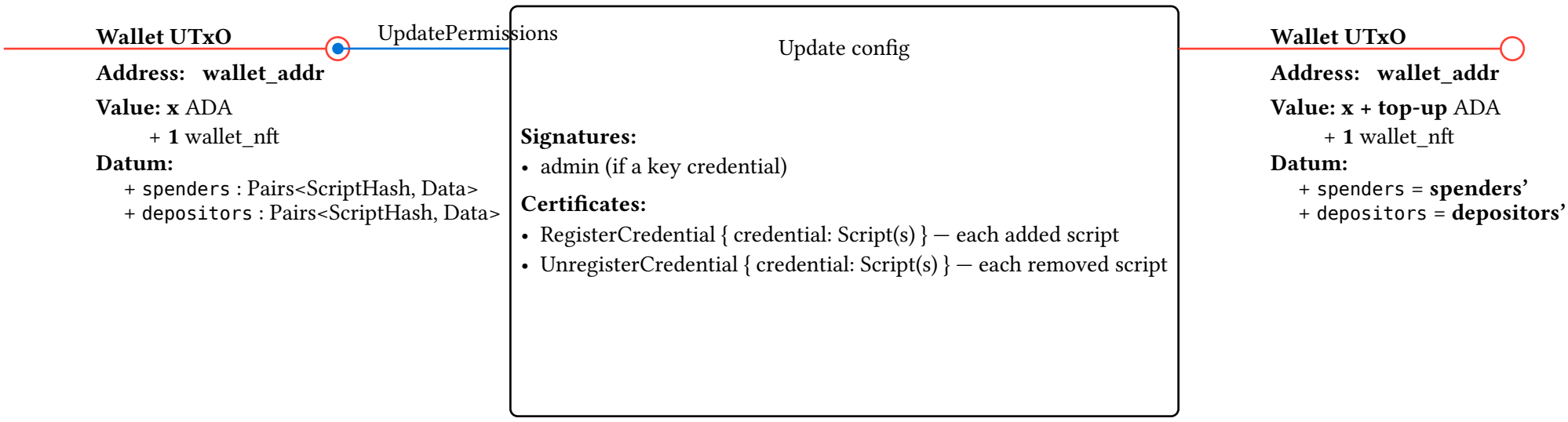
Notes:

- `wallet_addr`: payment is `wallet_hash`, staking any. A wallet UTxO holds its identifying NFT plus ada and arbitrary native assets.
- `wallet_hash = hash(smart_wallet(seed_utxo, settings_ref, wallet_nft_name, admin))`: the spending script is parameterized by its seed, configuration reference, NFT name, and the `admin` credential, so all wallet UTxOs share one address. Externalizing the members/threshold config lets it change without redeploying the script.
- validity: the spend requires the wallet UTxO to carry its NFT (`1` `wallet_nft` under `wallet_hash`) — only NFT-bearing UTxOs are valid smart wallets.
- base M-of-N floor: the wallet's configuration supplies `members` and `threshold`, and the spend requires at least `threshold` of `members` present in `extra_signatories`. `members` is treated as a set (duplicates count once), and the threshold must satisfy `0 < threshold <= distinct members`. Members are verification key hashes only in this version; script-credential members are not implemented. This design is agnostic of how that configuration is sourced.
- delegated authorizations: every `ScriptHash` key in the wallet UTxO's `spenders` datum field is a withdraw-0 staking script that must run in this transaction; the `Data` value carries that script's own mutable state. Keys in both delegated maps are unique. Stateless scripts store the canonical marker `0` and must keep it on every spend/deposit; stateful scripts enforce their own transition. The same hash may appear in both maps, but reference validators are single-role and their `publish` handler validates only the map it reads; prefer separate instances. The `Spend` redeemer asserts the conjunction — the M-of-N floor **and** all delegated checks — so no restriction can be silently omitted. The map is per-UTxO, so different wallet UTxOs may carry different restriction sets.
- the delegated scripts are the extension point: spending limits, time locks, recipient allow-lists, oracle attestations, … each is an independent withdrawal script observed here, none known to the spending script.
- wallet survival: the transaction must recreate the wallet — the only output at the wallet address must be NFT-bearing, carry the inline datum, and have a value that is a subset of the input's (a spend only removes value; additions belong to `Deposit`). Any NFT-less output at the wallet address is rejected, since it would be unspendable forever. Its `depositors` map must equal the input's and the `spenders` key set must stay fixed (only `UpdatePermissions` may change either), while spender `Data` may change: each spender script enforces its own transition. The NFT can therefore only leave the wallet on `Close`.
- composability: the spend asserts only its own input, the required spenders, and the surviving wallet with unchanged `depositors`, a fixed `spenders` key set, and a value no greater than the input's — never the total input or output counts, nor unrelated UTxOs. Payout outputs elsewhere are otherwise unconstrained.

Figure 2: Spend (base M-of-N + delegated authorizations)

3 - Update config (per-UTxO)

An authorized admin credential — a parameter of the spending script — rewrites a wallet UTxO's config (`spenders` and `depositors`) in place. Non-ADA assets are untouched: same address, same tokens — only the config changes. ADA may be topped up explicitly.



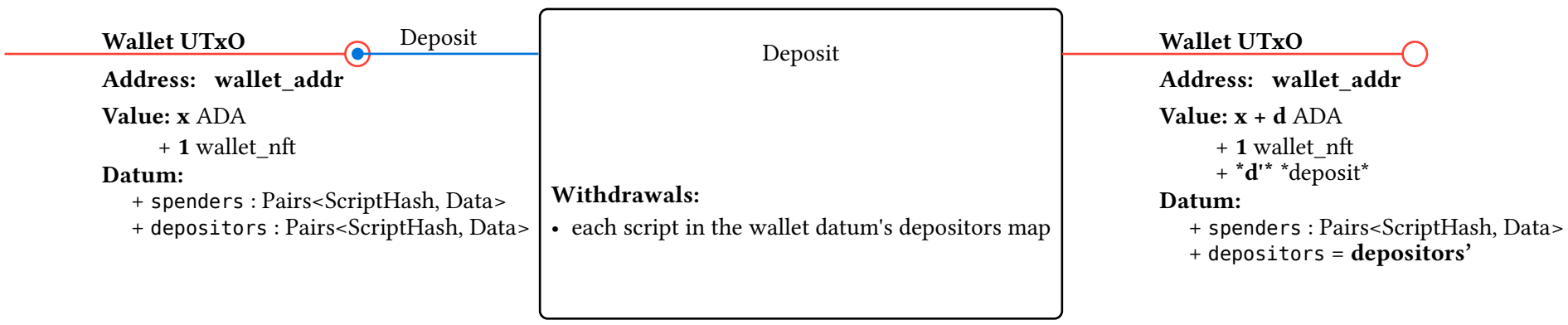
Notes:

- the `admin` credential (a script parameter) must be satisfied: a key admin signs; a script admin authorizes by a withdraw-0 invocation (the same pluggable mechanism used elsewhere) and must implement a `withdraw` handler. The reference `smart_wallet` validator is spend/mint-only, so it cannot administer another wallet.
- the continuation must sit at the same `wallet_addr` and preserve every non-ADA asset exactly, including the NFT. Lovelace may only increase: this explicit top-up lets the admin cover a larger datum/min-UTxO but bypasses the deposit scripts for that ADA. Apart from the top-up, an update cannot move funds, only swap the config. The admin is fixed in the script parameter, so control cannot be handed off.
- `spenders'` / `depositors'`: the new config. It takes effect immediately for subsequent spends and deposits of this UTxO; other wallet UTxOs are unaffected (the config is per-UTxO).
- the update diffs the old and new maps (both `spenders` and `depositors`): every script **added** must be registered here (`RegisterCredential` and its matching `Publish` redeemer, running its `publish` handler, which reads the script's initial `Data` from the continuation datum and validates it); every script **removed** must be unregistered (`UnregisterCredential`, whose `publish` handler only accepts the unregister when it sees the script leaving the wallet); every script **kept** must carry unchanged `Data`. This ensures each entry's `Data` was self-validated at registration and stays coherent across updates. Unregistering and re-registering the same hash in one transaction is still a **kept** entry, so its `Data` cannot change; re-initialize it across two transactions instead. A credential registered before this transaction cannot be added: the ledger rejects a second registration, so it must be unregistered first.

Figure 3: Update config (per-UTxO)

4 - Deposit (depositors validate added funds)

The wallet's `depositors` scripts validate funds added to the wallet UTxO — the UTxO is spent and recreated at the same address with the NFT, `spenders` and the `depositors` key set preserved, and no fund removed. Every deposit script must run and approve; with none configured, a deposit is open. A deposit cannot spend funds or change the permission set.



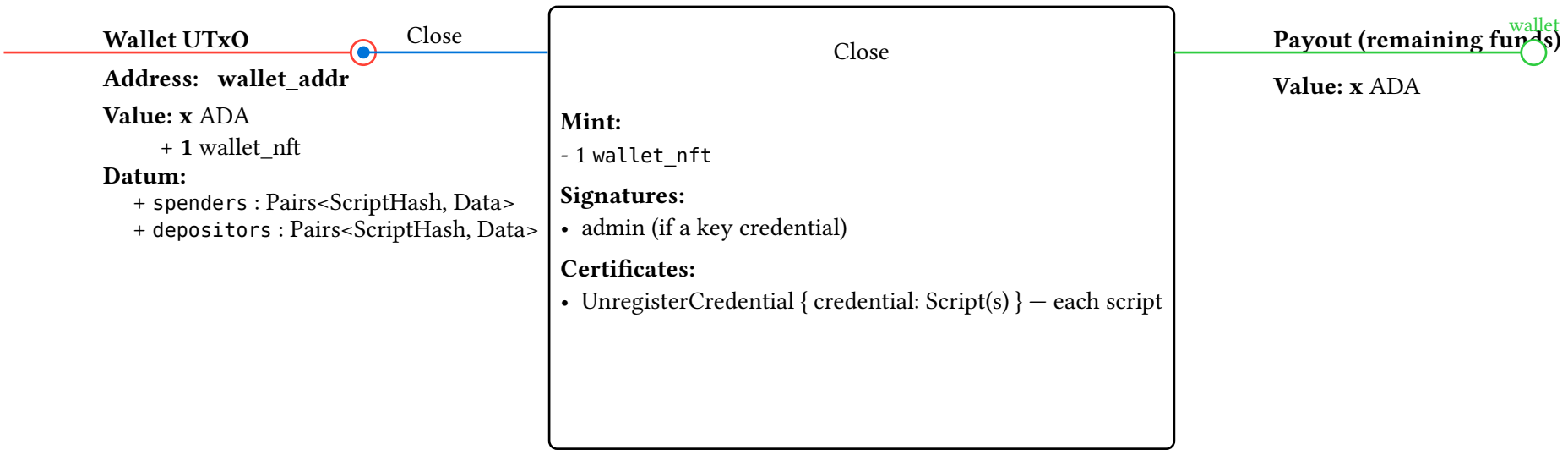
Notes:

- every script in the wallet UTxO's `depositors` datum field is a withdraw-0 staking script that must run and approve in this transaction — e.g. asset allow/denylists, source-address checks, or a key check. With an empty map, a deposit needs no approval.
- the continuation must sit at the same `wallet_addr` and preserve `spenders` and the `depositors` key set; depositor `Data` may advance (`depositors'`), with each deposit script enforcing its own input→output transition. A deposit cannot add or remove deposit scripts, or touch the spend side.
- the continuation's value must be a superset of the input's: every asset (including lovelace and the NFT) is present in at least the same quantity, so a deposit can only add funds, never spend or remove them.
- the `spenders` scripts do not run on a deposit: there is no outflow to restrict.

Figure 4: Deposit (depositors validate added funds)

5 - Close (admin burns the NFT)

The admin closes the wallet: the wallet UTxO is spent, releasing whatever funds remain, and the identifying NFT is burned. The `Close` spend redeemer requires the admin, the burn, and that every delegated script (spender and deposit) is unregistered so its stake deposit is refunded; the `Burn` mint redeemer permits the burn.



Notes:

- the `admin` credential (a script parameter) must be satisfied: a key admin signs; a script admin authorizes by a withdraw-0 invocation.
- the wallet NFT — `1 wallet_nft` under `wallet_hash` — must be burned in this transaction (`-1`), destroying the wallet's identity. The burn runs the mint endpoint with a `Burn` redeemer, which only checks the burn quantity; authorization is enforced by the `Close` spend redeemer.
- closing spends the wallet UTxO and releases any remaining funds. The admin may send them anywhere: `Close` does not enforce the M-of-N floor or any delegated spender/depositor restriction. There is no continuation: once the NFT is gone, the wallet can no longer be spent.
- every script in the closing wallet's `spenders` and `depositors` maps must be unregistered here (`UnregisterCredential`), refunding its stake deposit. Each script's `publish` handler accepts this because it sees itself in the wallet input and no wallet output.

Figure 5: Close (admin burns the NFT)

6 - Example: stateful spending window (withdrawal script)

A stateful delegated withdrawal script that limits the wallet to at most one `Spend` per window. Unlike `spending_limit` (whose `bound` is fixed at compile time), this script carries mutable state — the upper bound of the previous spend's validity range — in the wallet datum's `spenders` map, and enforces its own state transition on every spend.

The script is parameterized by `wallet`, the wallet NFT's asset name, and `window` (milliseconds), and its state is a single timestamp: `SpendingWindowState { last_upper: Int }` (POSIX ms; `0` means never spent).

- **Publishing** (`RegisterCredential`): the `publish` handler reads its initial `Data` from the NFT-bearing wallet output datum and requires `last_upper == 0` — an unspent wallet. Unregistering is only accepted while leaving the wallet (the same NFT-bearing input/output presence check as `spending_limit`).
- **Withdraw** (runs on every `Spend`): the handler requires both validity bounds to be finite, reads its current state from the NFT-bearing wallet **input** datum and the next state from the NFT-bearing wallet **output** datum, and requires the cooldown elapsed (`lower - last_upper >= window`, or it is the first spend) and that the output state records this spend's upper bound (`next.last_upper == upper`). Because execution happens between the lower and upper bounds, storing the upper bound makes the cooldown conservative even if the lower bound is backdated.
- Because the wallet's `Spend` branch doesn't constrain the continuation datum, the script enforces its own transition end-to-end: the builder must produce a continuation carrying the updated state, or the withdrawal fails and the transaction is rejected.

This is the template for any stateful restriction (spending budgets, time locks, allow-lists that mutate, …): the state lives in `spenders[own_hash]`, the `publish` handler validates the initial value, and the `withdraw` handler reads the input state, validates the transition, and checks the output state.